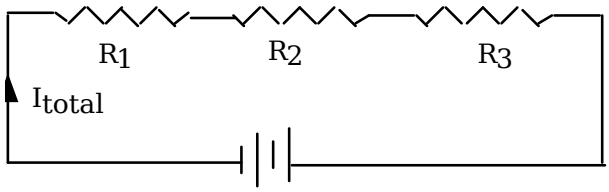
SERIES AND PARALLEL CIRCUITS

There are two basic arrangements of components in a circuit.

Series Circuits

The resistors are added end-to-end so that the ***whole current flows through each resistor*** in turn.

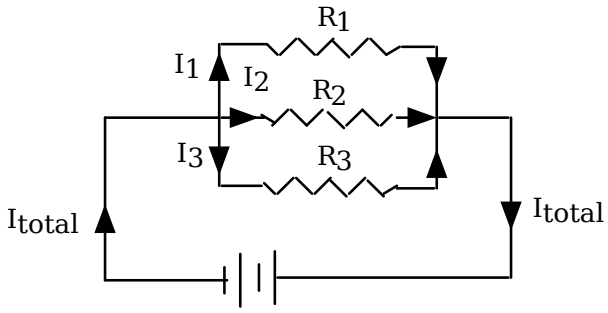


THE SAME CURRENT FLOWS THROUGH EACH RESISTOR.

- There is only ***one path*** for the current through the circuit.
- If one component stops working, they all stop working.
- ***Resistance increases*** with more components added.

Parallel Circuits

The resistors are on separate branches, and the ***current splits up*** to flow through each branch, depending on its resistance.



NOTICE HOW THE CURRENT I_{total} SPLITS UP INTO I_1, I_2 AND I_3 AS IT ENTERS THE PARALLEL PART OF THE CIRCUIT.

- ***Current splits up*** - there are multiple pathways.
- If one component is turned off, the others keep working.
- ***Resistance decreases*** for more and more resistors connected in parallel.